

GF(27) in GALG and FFGFT:

Why $G(3)$ Contains No GF(27) but $G(6)$ Does

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1 September 2026

Contents

GF(27) in GALG and FFGFT	2
.1 Background and Notation	1
.2 Occasion and Starting Point	2
.3 Theorem A: $G(3)$ Contains No GF(27)	2
Structure of $G(3)$	2
Eigenvalue Structure	2
Consistency with Doug's Findings	3
.4 Theorem B: $G(6)$ Contains Order-26 Elements	3
Structure of $G(6)$	3
Construction via Irreducible Polynomial	3
Embedding in $G(6)$ as a GALG Element	3
Minimal Example: 5 Blades, Directly Verifiable in GALG	3
Frequency in $G(6)$	4
.5 Theorem C: GF(27) as Order Structure, not Subfield	4
Subfield Criterion	4
Connection to Docs. 330/338	4
.6 Vacuum Structure and Z3 Separation (pruef_341_vakuum_witt_z3)	5
Witt Pairs and the Number Operator N	5
Triality Built into Z3C	5
Z3 Separation of Even and Odd Vacuums	5
Bilaterals as Sector-Changers (Gluon Role)	6
Projector Properties	6
.7 Consequences for FFGFT and GALG	6
Appendix: For the Reader Without an Algebra Background	6

GF(27) in GALG and FFGFT

Abstract

Doug Matzke reports in his IPI mail of 1 September 2026 that a systematic search in GALG $G(3)$ (6561 attempts) finds no element of order 26. This document explains algebraically why, shows where such elements exist in $G(6)$, and draws the consequences for the FFGFT mass layer.

Three theorems: **(A)** $G(3) \cong M_2(\text{GF}(9))^2$: order 26 is algebraically impossible in $G(3)$ — not a search error, but a theorem. **(B)** $G(6) \cong M_8(\text{GF}(9))$: order-26 elements exist; a 5-blade example can be verified directly in GALG. **(C)** $\text{GF}(27)$ is present in $G(6)$ as an order structure, not as a subfield (since $3 \nmid 8$) — exactly the FFGFT statement from Doc. 338.

Consequence: $1/\alpha = 3700/27$ lives on the $G(3)/\text{GF}(9)$ level; the lepton mass layer $(m_\mu/m_e)^2 = 43200$ needs $G(6)$ with order-26 elements. Verification script: `pruef_341_gf27_in_galg.py` (all assertions **[B]**).

.1 Background and Notation

This section provides the necessary background for readers not immediately familiar with all the algebraic structures involved. It can be skipped by specialists.

Status markers

This document uses the standard FFGFT corpus status markers:

- **[K]** — *Confirmed*: algebraically or numerically fully proved; all assertions in the verification script passed.
- **[B]** — *Established*: numerically secured by sampling or explicit construction; analytic proof present or straightforward.
- **[S]** — *Speculative*: indication or conjecture, not yet fully proved.

Galois fields $\text{GF}(p^n)$

A *Galois field* $\text{GF}(p^n)$ is the unique finite field with p^n elements, where p is prime (the *characteristic*) and $n \geq 1$. Its multiplicative group $\text{GF}(p^n)^* = \text{GF}(p^n) \setminus \{0\}$ is cyclic of order $p^n - 1$.

- $\text{GF}(3)$: the integers $\{0, 1, 2\}$ modulo 3. Multiplicative group order 2.
- $\text{GF}(9) = \text{GF}(3^2)$: elements $a + bi$ with $a, b \in \{0, 1, 2\}$ and $i^2 = -1 \equiv 2$. Multiplicative group order 8. This is Doug's symmetric Z3C.
- $\text{GF}(27) = \text{GF}(3^3)$: 27 elements, multiplicative group order $26 = 2 \cdot 13$.
- $\text{GF}(81) = \text{GF}(3^4)$: 81 elements, multiplicative group order $80 = 2^4 \cdot 5$.

The *order* of an element x in a group is the smallest positive k with $x^k = 1$. If the group has order m , then every element order divides m (Lagrange's theorem). This is the key fact used throughout.

Clifford algebra $\text{Cl}(n)$ and $\text{GALG } \mathbf{G}(n)$

The Clifford algebra $\text{Cl}(n)$ is generated by n basis vectors $e_1, \dots, e_n$ with the rule $e_i^2 = +1$ (hyperbits in Matzke's language) and $e_i e_j = -e_j e_i$ for $i \neq j$. A *blade* of grade r is a product $e_{i_1} e_{i_2} \dots e_{i_r}$ with $i_1 < i_2 < \dots < i_r$; together with the scalar 1 (grade 0) they form a basis of 2^n elements. $\text{GALG } \mathbf{G}(n)$ is $\text{Cl}(n)$ over the coefficient field $\text{GF}(9) = \text{Z3C}$ (i.e. each blade carries a $\text{Z3C} = \text{GF}(9)$ coefficient). As a matrix algebra: $\text{Cl}(3) \cong M_2(\text{GF}(9))^2$ and $\text{Cl}(6) \cong M_8(\text{GF}(9))$, where $M_k(F)$ denotes $k \times k$ matrices over the field F .

Why $\text{GF}(27)$ and order 26?

The lepton mass ratio $(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$ involves $27 = |\text{GF}(27)|$ (Doc. 338, pruef_330/331). The multiplicative group $\text{GF}(27)^*$ has order $26 = 2 \cdot 13$. An element of order 26 in any algebra generates a copy of $\text{GF}(27)^*$. Doug's search for order-26 elements in $\mathbf{G}(3)$ is therefore a direct test of whether $\text{GF}(27)$ structure is present in $\mathbf{G}(3)$.

Companion matrix

Given a monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$, its *companion matrix* is the $n \times n$ matrix with 1's on the subdiagonal and the coefficients $-a_0, \dots, -a_{n-1}$ in the last column. Its

characteristic polynomial is $f(x)$, so its eigenvalues are the roots of f and its order in GL_n equals the multiplicative order of the roots.

.2 Occasion and Starting Point

Doug Matzke (IPI, 1 Sept. 2026) reports finding in GALG $G(3)$: $GF(9)$ in the generator $(a+ja)^8 = 1$; $GF(81)$ in $G(3)$ with $(1+a+b+c+(a \wedge b))^{80} = 1$; and element orders $\{2, 3, 4, 8, 16, 32, 40, 80\}$ — all of which divide $80 = |GF(81)^*|$, consistent with the $Cl(3) \cong M_2(GF(9))^2$ structure where eigenvalues live in $GF(81)$.

An explicit search for elements of order 26 (which would signal $GF(27)^*$ structure):

`gasolve([a,b,c], is_root26): Attempted 6561 with 0 found.`

$6561 = 3^8$ is the number of elements in a 4-trit search space — exhaustive at that depth.

This is not a search error or a limitation of the tool. It is an algebraic theorem: order 26 requires the prime factor 13, and 13 does not appear in $80 = |GF(81)^*|$. The following sections prove this precisely and show where $GF(27)$ structure does appear — in $G(6)$, not $G(3)$.

.3 Theorem A: $G(3)$ Contains No $GF(27)$

Structure of $G(3)$

GALG $G(n) = Cl(n)$ over the symmetric $Z3C = GF(9) = GF(3)[i]$, where $i^2 = -1$, characteristic 3. For odd $n = 3$:

$$G(3) = Cl(3) \cong M_2(GF(9)) \oplus M_2(GF(9)) \quad (1)$$

Every irreducible representation is 2-dimensional over $GF(9)$.

Eigenvalue Structure

The eigenvalues of a 2×2 matrix over $GF(9)$ lie in the algebraic closure — in the smallest extension field $GF(9^2) = GF(81)$.

$$|GF(81)^*| = 80 = 2^4 \cdot 5 \quad (2)$$

The number 13 does not divide 80. Therefore:

- No element of $M_2(GF(9))$ has an order divisible by 13.
- Order 26 requires $13 \mid \text{ord}(X)$ — impossible in $G(3)$.
- Doug's search space $\{a, b, c\}$ spans $G(3)$ — the result 0/6561 is algebraically forced **[B]**.

Consistency with Doug's Findings

Table 1: Orders in GALG $G(3)$ and their Galois origin

Order	Origin	Divides
2	$ \text{GF}(3)^* $	80
4	$\varphi(5) = 4$	80
5	min. prime factor $ \text{GF}(81)^* $	80
8	$ \text{GF}(9)^* $	80
16, 20, 40, 80	subgroups of $\text{GF}(81)^*$	80
3	unipotent (char. 3)	—
26	$ \text{GF}(27)^* $, needs factor 13	not 80

A numerical sample of 3000 random $\text{Cl}(3)$ elements confirms: no order is divisible by 13 **[B]**.

.4 Theorem B: $G(6)$ Contains Order-26 Elements

Structure of $G(6)$

$$G(6) = \text{Cl}(6) \cong M_8(\text{GF}(9)) \quad (3)$$

Irreducible representations are 8-dimensional over $\text{GF}(9)$.

Construction via Irreducible Polynomial

The polynomial $f(x) = x^3 + 2x + 1$ over $\text{GF}(3)$ (equivalent to $x^3 - x + 1$) has no root in $\text{GF}(3)$ and is therefore irreducible. Since $\gcd(3, 2) = 1$, f remains irreducible over $\text{GF}(9)$; its roots lie in $\text{GF}(27) \setminus \text{GF}(3)$. The companion matrix

$$C_f = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\text{GF}(3)) \quad (4)$$

has order 26 in $\text{GL}_3(\text{GF}(3))$ (f is primitive) **[B]**.

Embedding in $G(6)$ as a GALG Element

The 8×8 element $X = C_f \oplus C_f \oplus (-I_2)$ satisfies:

- $X^{26} = 1$, $X^{13} \neq 1$, $X^2 \neq 1$ — order exactly 26 **[B]**
- Blade decomposition over 38 Z3C blades from $\{e_1, \dots, e_6\}$

Minimal Example: 5 Blades, Directly Verifiable in GALG

A systematic search shows: order-26 elements need at least 5 blades; with 4 blades there are none. The following 5-blade element can be verified directly in Doug's GALG tool:

$$X = -(1+i)(e_1e_2) + (1+i)(e_1e_2e_3e_6) + (1-i)(e_1e_2e_5e_6) + (1+i)(e_1e_4e_5e_6) - (e_1e_2e_3e_4e_6) \quad (5)$$

GALG syntax:

```

X = -(1+1j)*(e1^e2) + (1+1j)*(e1^e2^e3^e6)
+ (1-1j)*(e1^e2^e5^e6) + (1+1j)*(e1^e4^e5^e6)
- (e1^e2^e3^e4^e6)
X**26 == 1 # True
X**13 == 1 # False

```

[B] (pruef_341, all assertions passed)

Frequency in $G(6)$

Table 2: Fraction of elements with $13 \mid \text{ord}(X)$ in $G(6)$ by blade support

Support (number of blades)	Hits / Invertible	Fraction
≤ 4	0 / ~450	0 %
5	rare	< 1 %
6	85 / 478	17.8 %
10	190 / 511	37.2 %

Order-26 elements are common in $G(6)$ — Doug's search in $G(3)$ ($\text{span}\{a, b, c\}$) could not find them for two reasons: algebraically (Theorem A) and because $G(3) \subsetneq G(6)$.

.5 Theorem C: $\text{GF}(27)$ as Order Structure, not Subfield

Subfield Criterion

A unital embedding $\text{GF}(27) \hookrightarrow M_n(\text{GF}(9))$ makes $\text{GF}(9)^n$ a $\text{GF}(27) \otimes \text{GF}(9) = \text{GF}(729)$ -vector space. Since $[\text{GF}(729) : \text{GF}(9)] = 3$, we need $3 \mid n$.

Table 3: $\text{GF}(27)$ as subfield in $M_n(\text{GF}(9))$

n	$3 \mid n$	Subfield possible?
2	no	no ($G(3)$)
8	no	no ($G(6)$)
3	yes	yes
6	yes	yes
9	yes	yes

For $G(6)$ with $n = 8$: no $\text{GF}(27)$ subfield. But the minimal polynomial of X factors:

$$\chi_X(x) = (x^3 + 2x + 1)^2 \cdot (x + 1)^2 \in \text{GF}(3)[x] \quad (6)$$

Two cubic factors — eigenvalues in $\text{GF}(27)$ — and one linear factor from $\text{GF}(3)$. $\text{GF}(27)$ appears as splitting field, not as subalgebra.

Connection to Docs. 330/338

This is exactly the FFGFT statement from Docs. 330 and 338:

$$\text{GF}(9) \not\subset \text{GF}(27) \quad (\text{since } \gcd(2, 3) = 1) \quad (7)$$

Both fields sit in parallel in $\text{GF}(3^6) = \text{GF}(729)$, not in a tower. The mass-ratio identity $(m_\mu/m_e)^2 = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|$ uses both fields *in parallel* — consistent with Theorem C.

.6 Vacuum Structure and Z3 Separation (pruef_341_vakuum_witt_z3)

Doug's IPI mail of 1 September 2026 (22:05) not only verifies the order-26 element but discloses the full structure of six vacuum states $Vss[0 \dots 5]$ in $G(6)$ and their grade-2 (bivector) content. The following results are verified in `pruef_341_vakuum_witt_z3.py`.

Witt Pairs and the Number Operator N

The three grade-2 bivectors

$$B_1 = -i(e_1e_3), \quad B_2 = -i(e_2e_6), \quad B_3 = -i(e_4e_5) \quad (8)$$

satisfy $B_k^2 = +1$, commute pairwise, and commute with jE_7 . They form the three *Witt pairs* $\{13\}, \{26\}, \{45\}$ — the decomposition of $Cl(6)$ into three complex planes (Fock space of 3 modes). The number operator $N = B_1 + B_2 + B_3$ is simultaneously the grade-2 part of $Vss[0]$ and its constructor **[B]**:

$$(1 + N)(1 \pm jE_7) = Vss[0] \quad (\text{sign depends on basis orientation of } E_7) \quad (9)$$

Triality Built into Z3C

Over $\mathbb{C}$, the Fock number operator $N = B_1 + B_2 + B_3$ would have eigenvalues $\{-3, -1, +1, +3\}$. Over GF(9) in characteristic 3: $-3 \equiv 0 \equiv +3$ **[B]**.

$$N \bmod 3 \in \{0, +1, -1\} \quad \text{with multiplicities} \quad \{2, 3, 3\} \quad (10)$$

The vacuum ($n = 0$) and the fully occupied state ($n = 3$) *coincide* in characteristic 3. The $\mathbb{Z}_3$ cycle is closed without any extra input. This is exactly the FFGFT sector variable $\Delta_w = N \bmod 3$ (Doc. 336, $Q_{\mathbb{Z}_3} = N \bmod 3$). The +1-eigenspace is 3-dimensional — Doug's fermion-generation candidate.

Z3 Separation of Even and Odd Vacuums

The cyclic Witt-pair rotation

$$\sigma : \quad e_1 \rightarrow e_2 \rightarrow e_4 \rightarrow e_1, \quad e_3 \rightarrow e_6 \rightarrow e_5 \rightarrow e_3 \quad (11)$$

cycles $B_1 \rightarrow B_2 \rightarrow B_3 \rightarrow B_1$ while leaving E_7 invariant (even permutation). Applied to Doug's six vacuums **[B]**:

$$\begin{aligned} \sigma(Vss[0]) &= Vss[0], & \sigma(Vss[2]) &= Vss[2], & \sigma(Vss[4]) &= Vss[4] \\ \sigma(Vss[1]) &= Vss[3], & \sigma(Vss[3]) &= Vss[5], & \sigma(Vss[5]) &= Vss[1] \end{aligned} \quad (12)$$

Even vacuums $[0, 2, 4]$: individual $\mathbb{Z}_3$ -fixed points.

Odd vacuums $[1, 3, 5]$: a single $\mathbb{Z}_3$ -orbit of three.

This algebraic separation is the *same* as the Frobenius fixed-point/orbit split from Doc. 339: fixed points = massive sector (fermion generations), orbit = other sector (Doug's dark energy/mass candidate). Doug finds this split via grade parity; FFGFT finds it via Frobenius eigenstructure. **They agree [B]**.

Bilaterals as Sector-Changers (Gluon Role)

The Witt raisers $ap_k = (e_a + i e_b)/2$ for each pair satisfy **[B]**:

$$[N, ap_i \cdot ap_j] = +1 \cdot (ap_i \cdot ap_j) \quad (13)$$

The bilaterals $ap_i \cdot ap_j$ shift the N -sector by $\pm 1 \bmod 3$, i.e. they are Δ_w -changers. This is the gluon role in Doc. 339 (three-fold orbit transmutation between sectors). Doug's question "how do gluons transmute quarks using bilaterals?" is answered: they shift the winding-mode sector Δ_w by ± 1 .

Projector Properties

$$Vss[0]^2 = -Vss[0], \quad N \cdot Vss[0] = 0 \quad (14)$$

$-Vss[0] = 2 \cdot Vss[0]$ is idempotent over $GF(9)$; $Vss[0]$ projects onto the $N = 0$ sector (vacuum + fully occupied state).

.7 Consequences for FFGFT and GALG

Table 4: $GF(27)$ structure and vacuum separation in GALG and FFGFT

Level	GALG	FFGFT	Status
α	$G(3)/GF(9)$	$1/\alpha = 3700/27$, cancels	[B]
Lepton masses	$G(6)$, ord. 26 needed	$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$	[K]
$GF(27)$ subfield	absent in $G(3)$, $G(6)$	parallel in $GF(729)$	[B]
Vacuum split	$Vss[0, 2, 4]$ vs. $Vss[1, 3, 5]$	massive vs. other sector	[B]
$\mathbb{Z}_3$ structure	fixed pts vs. orbit	Frobenius split (Doc. 339)	[B]
N eigenvalues	$\{0, +1, -1\}$, mult. $\{2, 3, 3\}$	$\Delta_w \in \{0, +1, -1\}$	[B]
Bilaterals	$[N, ap_i ap_j] = +1$	Δ_w -changer, gluon role	[B]

The fine-structure constant $1/\alpha = 3700/27$ lives on the $GF(9)/G(3)$ level: $|GF(27)| = 27$ cancels in the derivation. The lepton mass layer and the vacuum structure both need $G(6)$. Doug's two-family split of vacuums and FFGFT's Frobenius separation find the same $\mathbb{Z}_3$ orbit structure from different starting points **[B]**.

Appendix: For the Reader Without an Algebra Background

This appendix explains the key ideas in plain language, assuming no prior knowledge.

What is a finite field?

A field is a set of numbers in which you can add, subtract, multiply and divide (except by zero). Familiar examples are the real numbers $\mathbb{R}$ and the rational numbers $\mathbb{Q}$. A *finite* field has only finitely many elements. You compute "modulo" a prime p : discard all multiples of p . Example $GF(3)$: the numbers $\{0, 1, 2\}$ with $1 + 2 = 0$, $2 + 2 = 1$, etc.

What does $\text{GF}(p^n)$ and $\text{GF}(p^n)^*$ mean?

$\text{GF}(p^n)$ is the unique finite field with exactly p^n elements. The *star* $\text{GF}(p^n)^*$ denotes all elements *except zero* — analogous to $\mathbb{Z}^* = \{\dots, -1, +1, \dots\}$ vs. $\mathbb{Z}$:

$$|\text{GF}(27)| = 27 \quad |\text{GF}(27)^*| = 26 = 27 - 1$$

Group order and Lagrange's theorem

$\text{GF}(p^n)^*$ is always a cyclic group of order $p^n - 1$. The *order* of an element x is the smallest k with $x^k = 1$. By Lagrange's theorem, every element order *divides* $p^n - 1$. Think of a clock hand that returns every 4 hours: it cannot fit on a 6-hour clock because 4 does not divide 6.

Concretely:

- $\text{GF}(81)^*$: order $80 = 2^4 \cdot 5$. Prime factors: 2 and 5 only. Order $26 = 2 \cdot 13$ is impossible — 13 does not divide 80.
- $\text{GF}(27)^*$: order $26 = 2 \cdot 13$. Elements of order 13 and 26 exist here.

Why does $\text{GF}(27)$ appear in the lepton masses?

$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 25 \cdot 27$. The 27 is $|\text{GF}(27)|$ (total field size including zero); the 8 is $|\text{GF}(9)^*|$ (group order of $\text{GF}(9)$). These arise because the FFGFT winding numbers on the T^4 torus are algebraically linked to these fields.

Why does Doug find no $\text{GF}(27)$ in $\text{G}(3)$?

$\text{G}(3)$ uses 2×2 matrices over $\text{GF}(9)$; eigenvalues live in $\text{GF}(81)$. $\text{GF}(81)^*$ has order 80, and $13 \nmid 80$, so order 26 is impossible. In $\text{G}(6)$ with 8×8 matrices, order 26 is achievable. A 5-blade example is given in §.4 and directly verifiable in GALG.

Verification Script

`pruef_341_gf27_in_galg.py` — Theorems A, B, C verified numerically ($\text{Cl}(6)$ over $\text{GF}(9)$ as 8×8 matrices; exact order determination; blade decomposition with Z3C coefficients; samples of 3000 and 600 elements). All assertions passed **[B]**.

`pruef_341_vakuum_witt_z3.py` — Vacuum structure verified: constructor identity, Witt pairs, N eigenvalues, $\mathbb{Z}_3$ separation of $\text{Vss}[0 \dots 5]$, bilaterals as sector-changers, projector properties. All assertions passed **[B]**.

Note on AI Assistance

This document was prepared with support from Claude (Anthropic). All algebraic computations were verified by Python scripts with exact arithmetic over $\text{GF}(9)$ as 8×8 matrix representations. The physical interpretations and conclusions are by Johann Pascher (ORCID 0009-0000-6518-4064).

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